

Multiple Choice (10 marks)

1. Which of the jovian planets have rings?
 - (a) Neptune
 - (b) Uranus
 - (c) Saturn
 - (d) all of the above
2. All of the following have been cited as evidence for the Giant Impact Hypothesis except:
 - (a) The Moon and Earth have nearly identical isotopic compositions.
 - (b) The Moon's orbit is not in the same plane as Earth's equator.
 - (c) The Moon's crust is thicker on the far side than the near side.
 - (d) The Moon has a much smaller iron core than the Earth even considering its size.
 - (e) The Moon was once entirely molten.
3. Across what potential difference in V does an electron have to be accelerated to reach the speed $v = 1.8 \times 10^7$ m/s? Calculate this relativistically.
 - (a) 750.0
 - (b) 924.0
 - (c) 1300.0
 - (d) 800.0
 - (e) 870.0
4. When a wave moves from shallow water to deep water, the
 - (a) frequency does not change, the wavelength increases, and the speed increases
 - (b) frequency decreases, the wavelength increases, and the speed does not change
 - (c) frequency increases, the wavelength increases, and the speed decreases
 - (d) frequency increases, the wavelength does not change, and the speed increases
 - (e) frequency increases, the wavelength decreases, and the speed does not change
5. Find the energy equivalent to 1 amu.
 - (a) 1000 Mev
 - (b) 650 MeV
 - (c) 800 MeV
 - (d) 931 Mev
 - (e) 500 Mev

True / False (10 marks)

Answer True or False

6. True or False: The energy of a photon with a wavelength of 400 nm is 1.9 eV.
7. True or False: The value of g in Paris is 33.000 ft/s^2 , which is higher than the value in London.
8. True or False: Sunspots are dark regions on the Sun's surface that appear in cycles approximately every 11 years, known as the solar cycle.
9. True or False: Plating a spoon at 0.50 ampere for 20 minutes results in a silver deposition of around 1.000 grams.
10. True or False: The time required for one revolution of a satellite is linear in the mass of the planet it orbits.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the scientific and technological advancements that enabled the U.S.S.R. to successfully send landers to Venus, and analyze their impact on our understanding of the planet.
12. A horizontal force of 5 N is applied to keep a 10 kg box sliding at a constant velocity of 2 m/s on a rough surface. Calculate the work done by this force over 1 minute and discuss the implications of friction in this scenario.

End of Paper